

Explain the difference between Java Persistence API, Hibernate and Spring Data JPA

Feature	JPA (Java Persistence API)	Hibernate	Spring Data JPA
Definition	A Java specification (JSR 338) for object-relational mapping	An ORM framework that implements JPA	A Spring framework built on top of JPA
Type	Specification	Implementation	Abstraction Layer
Purpose	Defines standard APIs for database persistence	Provides the actual implementation of JPA and ORM features	Simplifies database operations by reducing boilerplate code
Dependency	Cannot work alone	Can work independently or with JPA	Requires JPA and a JPA provider like Hibernate
Coding Effort	Moderate	Moderate	Least
Repository Support	No	No	Yes (JpaRepository)
SQL Generation	Defined by implementation	Automatically generates SQL	Uses Hibernate to generate SQL automatically